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Elastic Net Regression.

combination of both

Ridge Regression

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|w\|^2$$

Lasso Regression.

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \|w\|$$

Elastic Net: $L = \sum (y_i - \hat{y}_i)^2 + a \cdot \|w\|^2 + b\|w\|$

hyperparameter $\begin{cases} \lambda = a + b \\ l1\text{-ratio} = \dfrac{a}{a+b} \end{cases}$

default: $\lambda = 1$
$l1\text{-ratio} = 0.5$
$a = b = 0.5$

If $l1$-ratio $0.9 \rightarrow$ 90% ridge
10% lasso

- Use elastic net regression if input columns has multicolinearity.

```
from sklearn.~~learn~~linear_model import ElasticNet.
reg = ElasticNet(alpha=0.005, l1_ratio=0.5)
reg.fit(X_train, y_train)
y_pred = reg.predict(X_test)
r2_score(y_test, y_pred)
```

OR

SGDRegressor(penalty='elasticnet')